

Fair Price in a Moving Market: Predicting Residential Prices

Phase 2: Data Preparation, Cleaning and Feature Generation

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ABSTRACT

This report aims to explore data set preparation for our machine learning project. The initial data set chosen was in the context of Egyptian real estate prices, but after instructor feedback we replaced it with the Zillow Prize (Zestimate) property dataset from Kaggle, which fit the required parameters. After pre-processing and cleaning we obtained 12 features. We used statistical inference to clean data, replace missing values and scale features in order to obtain a clean dataset of 2,950,951 records with no missing values to use for a linear regression model. The label being predicted is `taxvaluedollarcnt`.

KEYWORDS

data cleaning, feature generation, encoding, missing values, scaling

1 DATASET SIZE

The original size was 2,985,217 rows and 58 columns. After removing the 34,266 rows with a missing label, we have 2,950,951 rows and 48 columns. These columns include 7 numeric and 5 categorical feature columns, in addition to all the one-hot encoding and frequency encoding columns.

2 DROPPED COLUMNS

2.1 Leakage

Table 1: Columns dropped because they leak the label.

Column	Why
<code>structuretaxvaluedollarcnt</code>	Addition with another column gives the label, so leaking occurs
<code>landtaxvaluedollarcnt</code>	Addition with another column gives the label, so leaking occurs
<code>taxamount</code>	Addition with another column gives the label, so leaking occurs

2.2 High missingness, and no way to find what the missing values represent

2.3 Redundant with a kept feature (13)

2.4 Not useful

3 KEPT FEATURES

3.1 Numeric

- `bathroomcnt`: low number of missing values, and the median rises with the number of bathrooms.
- `bedroomcnt`: a steady increase in value with increasing rooms is observed.
- `yearbuilt`: gives property age, so it has semantic importance.
- `calculatedfinishedsquarefeet`: a strong correlation is observed.
- `lotsizesquarefeet`: adds land-related semantic information.
- `latitude` and `longitude`: finer location parameters that work with `fips`.

3.2 Categorical

- `buildingqualitytypeid`: each quality shows up with a different value, so it does contribute.
- `fips`: geographical information is a big factor in real estate pricing.
- `heatingorsystemtypeid`: different values for different identifiers, so it does contribute.
- `propertylandusetypeid`: value differences are observed.
- `propertycountylandusecode`: a low number of missing values, and more detailed than `propertylandusetypeid`.

4 TRANSFORMATIONS

4.1 Categorical encodings

One-hot (4 features): `buildingqualitytypeid`, `fips`, `heatingorsystemtypeid` and `propertylandusetypeid`.

Table 2: Columns dropped for high missingness.

Subgroup	Columns	Why
Pool and spa (6)	poolcnt, poolsizesum, pooltypeid2, pooltypeid7, pooltypeid10, hashottuborspa	Close to 99% missing rows; inadequate pool information of any kind to build a feature
Fireplace, garage, deck (5)	fireplacecnt, fireplaceflag, garagecarcnt, garagetotalsqft, decktypeid	fireplacecnt 89.51% missing, garage columns above 70%; decktypeid 99.42% missing with a single observed category (66), so no variation
Building structure (11)	airconditioningtypeid, architecturalstyletypeid, buildingclasstypid, basementsqft, storytypeid, typeconstructiontypeid, threequarterbathnbr, numberofstories, yardbuildingsqft17, yardbuildingsqft26, unitcnt	basementsqft is 99.95% missing; threequarterbathnbr is also just one component of the bathroom total; unitcnt is not over the missingness threshold, but propertylandusetypeid already gives us the information needed and it has fewer missing rows. The other columns are over the threshold
Tax delinquency (2)	taxdelinquencyflag, taxdelinquencyyear	98.1% missing, and also removed for semantic reasons, as these columns have to do with the owner's behaviour
Zoning and census identifiers (2)	propertyzoningdesc, censustractandblock	propertyzoningdesc has thousands of codes, so encoding is difficult, and propertylandusetypeid and propertycountylandusecode are already sufficient; censustractandblock is a long geographic identifier, not a quantity, and location is covered by fips, latitude and longitude

Table 3: Columns dropped as redundant.

Column	Why
calculatedbathnbr	Follows bathroomcnt in 100% of cases, but it has more missing values
fullbathcnt	Follows bathroomcnt
finishedfloor1squarefeet, finishedsquarefeet12, 13, 15, 50, 6	Redundant with the feature calculatedfinishedsquarefeet, which is kept
rawcensustractandblock, regionidcity, regionidcounty, regionidneighborhood, regionidzip	Location is already represented by fips, latitude and longitude; the encodings would be large and would just show the same data

Table 4: Columns dropped for carrying no usable signal.

Column	Why
parcelid	A unique identifier does not contribute
assessmentyear	Most values are the same
roomcnt	77.6% recorded as zero rooms, so we use bathroom count and bedroom count as better indicators

The identifiers are just identifiers and the numbers themselves do not hold any actual weight, so one-hot encoding is used.

Frequency encoding (1 feature): the column propertycountylandusecode. With 234 categories, one-hot would add too many columns, so we use frequency encoding instead. The limitation with regard to collisions is noted.

4.2 Numeric transformations

- **Median imputation:** for bathroomcnt, bedroomcnt and yearbuilt.
- **Divide by 1,000,000, then median imputation:** latitude and longitude, because Zillow stores coordinates scaled by 10^6 .
- **Log, then median imputation:** calculatedfinishedsquarefeet and lotsizesquarefeet, because of large skewness.
- **Z-score on 8 numeric columns:** bathrooms, bedrooms, year built, log finished area, log lot size, latitude, longitude, and the frequency-encoded county code.

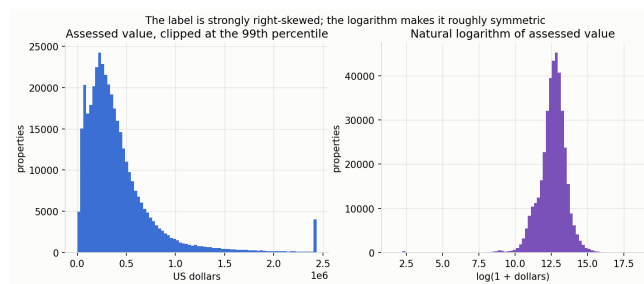
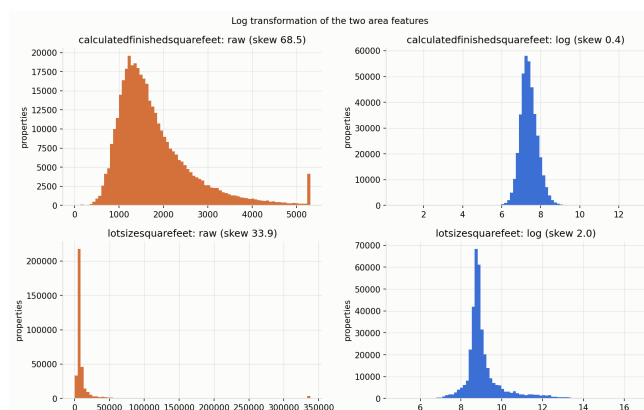
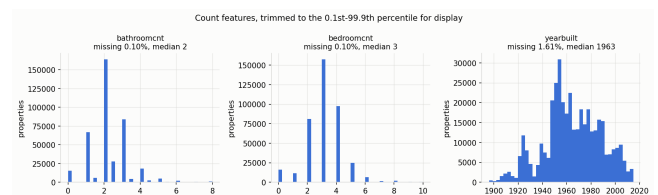
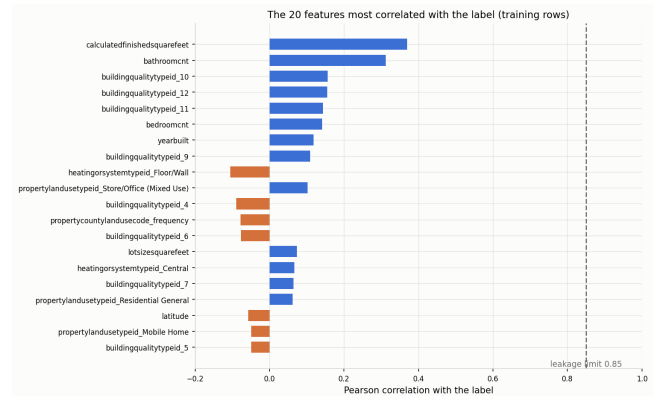
5 FIGURES

Table 5: The seven numeric features that were kept.

Feature	Missing	Unique	Min	Median	Mean	Max	Std	Skew	Correlation
bathroomcnt	0.099% (2,957)	39	0	2	2.22	32	1.08	1.18	0.308
bedroomcnt	0.099% (2,945)	25	0	3	3.09	25	1.27	0.50	0.138
yearbuilt	1.60% (47,833)	183	1801	1963	1964.4	2016	23.6	-0.15	0.130
calculatedfinishedsquarefeet	1.51%	12,631	1	1,574	1,831	952,576	1,954	156.4	0.560
lotssquarefeet	9.14% (272,706)	70,214	100	7,000	22,604	371,000,500	249,984	1,208.5	0.005
latitude (degrees)	0.098% (2,932)	852,718	33.324	34.008	34.001	34.820	0.244	0.28	-0.057
longitude (degrees)	0.098% (2,932)	1,043,238	-119.476	-118.172	-118.202	-117.554	0.346	-0.78	-0.020

Table 6: The five categorical features that were kept.

Feature	Missing	Unique	Largest categories	Encoding
buildingqualitytypeid	34.97% (1,043,822)	12	6 (561,502), 8 (501,925), 4 (448,049)	One-hot, missing as "Unknown"
fips	0.098% (2,932)	3	6037 (67.4%), 6059 (25.0%), 6111 (7.5%)	One-hot, missing as "Unknown"
heatingorsystemtypeid	37.39% (1,116,053)	14	2 (1,168,610), 7 (646,021)	Identifiers mapped to heating-system names, then one-hot, missing as "Unknown"
propertylandusetypeid	0.098% (2,932)	16 (15 after label removal)	261 (72.1%), 266 (16.2%)	One-hot, missing as "Unknown"
propertycountylandusecode	0.10% (2,999)	234	0100 (1,153,332), 122 (522,127), 0101 (248,386)	Frequency encoding, missing as "Unknown"

**Figure 2: The label before and after a logarithm.****Figure 3: The two area features before and after the logarithm, with the skew of each.****Figure 4: The three count features, with their missingness and median.****Figure 5: The strongest correlations with the label, against the 0.85 limit.**

